Counting bit strings

Theorem

Let n and r be integers with $0 \leq r \leq n$. Then

$$\binom{n+1}{r+1} = \sum_{j=r}^{n} \binom{j}{r}.$$

Recall: We can prove combinatorial identities by counting bit strings with different properties!

LHS is the number of bitstrings of length (n+1) that contain exactly (r+1) ones.